Problem Statement and Goals for Verity Software Engineering

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October 19, 2026

Table 1: Revision History

Date	Developer(s)	Project and Scope Change
2026-09-24	Calvin Salsali, Raymond Tao, Jaden Moore, Richard Han	Defined initial problem scope around digital content authenticity, inputs/outputs, and preliminary product goals (`G1`–`G6`).
2026-09-28	Calvin Salsali, Raymond Tao	Narrowed initial input modalities to text and static images, added custom documentation plan (Fall Machine Learning Model Report and Winter Usability Report), and defined ephemeral privacy goal (`G6`).
2026-10-19	Calvin Salsali, Raymond Tao, Jaden Moore, Richard Han	Addressed TA feedback (Issue #18, #20, #21) and Peer Review Team 12 feedback (Issues #14–#16): restructured Problem section into *Problem* → *What It Causes* → *What Verity Addresses*, reframed `G1`–`G6` as high-level product selling points, excluded opinion/satire from truth scoring, expanded stakeholder and persona analysis, and added individual member reflections and verified GenAI disclosure.
2026-10-19	Jaden Moore (Codex-assisted)	Distinguished the no-persistence privacy rule from the bounded cleanup of application-owned transient content.

1 Problem Statement

This section defines the problem addressed by Verity, its real-world consequences, the high-level inputs and outputs of the system, its primary and secondary stakeholders, and its operating environment.

1.1 Problem, Consequences, and What Verity Addresses

The Problem: People browsing the web increasingly encounter digital text and static images whose origin and authenticity are difficult to evaluate at a glance. Advances in generative Artificial Intelligence (AI), Large Language Models (LLMs), and digital image synthesis or splicing tools allow convincing reviews, articles, social posts, and photographs to be fabricated or altered at scale, while the technical signals that distinguish synthetic or manipulated media from authentic content remain hidden from everyday users.

What It Causes: This environment creates two symmetrically harmful failure modes for users and society. First, a reader who accepts synthetic or manipulated content at face value can suffer financial deception, professional or academic misjudgment, and contribute to the spread of deceptive media. Second, a reader who relies on opaque, single-score "real vs. fake" detectors can wrongly distrust authentic human writing or genuine photographs—causing reputational harm and unfair accusations against legitimate authors (particularly non-native English as a Second Language (ESL) writers whose formal prose triggers false positives in commercial detectors, and photographers whose images have had metadata stripped by social platforms).

What Verity Addresses: Verity addresses this problem by providing a user-triggered desktop browser extension and analysis pipeline that helps users evaluate suspicious webpage text and static images through inspectable evidence signals, explicit uncertainty and abstention (`inconclusive`) communication, and actionable independent verification steps. Rather than claiming to be an infallible arbiter of factual truth, political opinion, satire, or authorial intent (Peer Issue #15), Verity surfaces the concrete generation, manipulation, and provenance signals behind its assessment while processing selected items ephemerally with zero default storage.

1.2 Inputs, Outputs, and Scope Boundaries

High-Level Inputs:

- A user-selected webpage text excerpt or static image (PNG, JPEG, WebP) explicitly triggered for analysis by the user within supported size bounds.
- Optional publicly visible element context (such as image container metadata, Exchangeable Image File Format (EXIF) / Coalition for Content Provenance and Authenticity (C2PA) tags, or user-supplied context notes).

High-Level Outputs:

- A structured, non-accusatory signal assessment bound to the selected item, or an explicit `inconclusive` / `failed` status when signals are weak, conflicting, out-of-domain, or unavailable.
- A source-grounded summary of the specific signals observed in the input (distinguishing direct input signals, external references, and model inferences).
- An explicit indication of uncertainty and task limitations so the output is never mistaken for a calibrated probability of truth or fraud.
- At least one concrete next verification step (such as checking primary sources, provenance credentials, or reverse-image search) when evidence is incomplete.

Explicit Out-of-Scope Exclusions (Peer Issue #15): Verity evaluates bounded AI-generation, manipulation, and provenance signals; it does *not* score the "trustworthiness" of pure opinion, political commentary, satire, parody, or creative fiction, and it does *not* perform covert whole-page scraping or audio/video deepfake streaming analysis.

1.3 Stakeholders and User Profiles (Peer Issue #16)

Table 2: Stakeholder Analysis and Real-World Impact

Stakeholder Group	Role	Needs, Concerns, and Impact of Verity
Everyday Web Readers, Consumers & Older Adults (e.g., *Persona 1: Maya Lin, 34, Project Coordinator; older adults browsing marketplaces/news*)	Primary Hands-On Users	Encounter questionable reviews, marketplace listings, phishing scams, and viral images while browsing. Need low-friction in-page checks, plain-language explanations without ML jargon, and privacy protection for sensitive selections.
Students & Educators (e.g., *Persona 2: Marcus Vance, 21, Campus Copy-Editor*)	Primary / Secondary Users	Vet online sources and student submissions. Deeply concerned about commercial AI detectors falsely accusing ESL writers or penalizing benign editing, and about leaking unpublished drafts to third-party servers.
Journalists, Researchers & Community Moderators (e.g., *Persona 3: Priya Patel, 28, Forum Moderator*)	Secondary Power Users	Triage multiple flagged posts rapidly. Need clear separation between stripped EXIF/C2PA metadata and pixel-level forgery, explicit timeout/failure states, and strict isolation across concurrent checks.
Content Creators, ESL Authors & Depicted Persons	Indirectly Affected Stakeholders	Risk reputational damage or unfair dismissal if authentic writing or photography is falsely labelled "fake." Protected by Verity's non-accusatory framing, false-positive ceiling, and ESL/compressed-image subgroup fairness audits.
Society at Large, Course Staff & Team 23	Societal & Academic Stakeholders	Benefit from resilient media literacy, reduced automation bias, reproducible open-source benchmarks, and verifiable software engineering documentation.

1.4 Environment

The user runtime environment is a desktop Chromium-compatible web browser (Google Chrome, Microsoft Edge, or Brave supporting Manifest Version 3) on macOS, Windows, or Linux with standard Internet connectivity. End users require no specialized GPU hardware, local Python/CUDA installation, or machine learning expertise.

2 Goals (High-Level Product Selling Points)

In accordance with TA feedback (Issue #18), Goals **G1**–**G6** state the high-level product selling points and user value propositions of Verity in non-expert terms. Specific quantitative verification thresholds (such as Macro-$F_1 \geq 0.80$, FPR $\leq 10.0\%$, $+0.10$ baseline Macro-F_1 delta, 80% user comprehension, 10.0 s response latency, and 1.0 s application-owned transient cleanup with no persistent raw-content writes) are specified and justified in the companion Software Requirements Specification (SRS) [1].

G1. Evidence-First Assessment Over Opaque Labels: Verity gives users clear, inspectable reasons behind every content check rather than an unexplained binary "real" or "fake" score, allowing users to see which concrete text, visual, or metadata signals contributed to the assessment.

G2. Honest Uncertainty and Safe Abstention: Verity openly communicates when available evidence is weak, conflicting, out-of-scope (such as opinion or satire), or unavailable due to a service failure, returning an explicit `inconclusive` or `failed` status with concrete next verification steps rather than false reassurance.

G3. Defensible, Low-False-Accusation Detection Quality (Peer Issue #14): Verity delivers reliable detection on supported AI-generation and manipulation tasks while actively bounding false positives against legitimate human writing (including ESL and formal prose) and authentic photographs, backed by reproducible benchmark evaluation against naive baselines.

G4. Non-Expert Understandability and Non-Accusatory Framing: Verity translates technical forensic and statistical signals into plain-language summaries that everyday readers can understand without data-science training and without accusing authors of dishonesty or fraud.

G5. Seamless In-Page Browser Responsiveness: Verity lets users check suspicious text or images directly on the webpage they are viewing and returns results fast enough for interactive browsing without copy-pasting into external websites.

G6. Privacy-First Ephemeral Content Handling: Verity respects user confidentiality by accessing only user-selected webpage items, requiring explicit prior consent before any external provider transfer, and releasing application-owned transient copies within the SRS cleanup bound after every terminal outcome, without persisting raw content.

3 Stretch Goals

The following capabilities would add value beyond the core Revision 0 baseline if schedule slack permits:

- Richer multimodal cross-checking that correlates caption claims with image provenance manifests (`C2PA` Content Credentials visualizer).
- Expanded cross-browser packaging for Mozilla Firefox and Apple Safari Web Extensions in addition to desktop Chromium.

4 Custom Documents

- **Fall: Machine Learning Model Report.** Documents the evaluation methodology, benchmark datasets, baseline comparisons, calibration/abstention behaviour, and subgroup false-positive analysis used to validate Verity's detection pipeline.
- **Winter: Usability Report.** Evaluates whether representative non-expert users accurately interpret Verity's assessment cards, uncertainty indicators, task limitations, and verification guidance across completed, inconclusive, and failed scenarios.

Appendix A: Generative AI Use Disclosure

In accordance with the SFWRENG 4G06 Generative AI policy, this appendix summarizes the AI tools used in preparing this Problem Statement and the human verification steps completed by Team 23.

AI Tools Used

- **Google Gemini 3.1 Pro** (accessed October 2026 via Antigravity IDE assistant) — used to format the LaTeX tables and assist in editing the Problem Statement narrative flow and stakeholder table to address TA and Peer Review Team 12 feedback.
- **OpenAI Codex** (accessed October 2026) — used to assist in refining formal mathematical and privacy phrasing across the companion deliverables.
- **OpenAI ChatGPT (GPT-5)** (accessed September 2026) — used during initial September brainstorming to critique early draft wording against the course Problem Statement checklist.

How and Where Used

- **Calvin Salsali** and **Raymond Tao** used **Gemini 3.1 Pro** to restructure Section 1.1 (*Problem → What It Causes → What Verity Addresses*), format Stakeholder Table 2, and align Goals `G1`–`G6` with the companion SRS.
- **Jaden Moore** used **OpenAI Codex** to refine the ephemeral privacy phrasing in Goal `G6` (1.0 s bounded transient logical cleanup with zero persistent storage) to align with the companion SRS and Hazard Analysis.
- **Jaden Moore** and **Richard Han** used **Gemini 3.1 Pro** to verify LaTeX cross-references and synchronize the Revision History and reflection structure.

Completed Verification of AI-Generated Content

All AI-assisted edits were manually verified by the four members of Team 23 prior to repository commit:

1. **Scope and Goal Alignment Audit (Verified by Calvin Salsali & Raymond Tao):** Manually checked that Goals `G1`–`G6` contain high-level product selling points while all quantitative fit criteria are traced to `docs/SRS/SRS.tex`, and confirmed that opinion and satire are explicitly excluded (Peer Issue #15).
2. **Stakeholder, Fairness, and Citation Audit (Verified by Jaden Moore & Richard Han):** Verified all stakeholder groups, user personas, ESL false-positive bias claims, and PDF build output across `pdfs/ProblemStatementAndGoals/ProblemStatement.pdf`.
3. **Privacy Wording and Specification Consistency Audit (Verified by Jaden Moore & Raymond Tao):** Verified that Goal `G6` accurately specifies bounded transient cleanup (`MAX_RETENTION_TIME` = 1.0 s) with zero persistent raw storage, and confirmed that PDF formatting has no margin overruns or layout defects across `pdfs/ProblemStatementAndGoals/ProblemStatement.pdf`.

Appendix B: Reflection

1. **What went well while writing this deliverable?**
2. **What pain points did you experience during this deliverable, and how did you resolve them?**
3. **How did you and your team adjust the scope of your goals to ensure they are suitable for a Capstone project?**

Team Synthesis

Separating the user's trust decision problem from low-level implementation details helped us converge on a clear product vision: a user-triggered browser extension that provides evidence, uncertainty, and next verification steps rather than a black-box "truth" label. Incorporating feedback from Peer Review Team 12 (Issues #14–#16) and TA Rebecca Di Filippo (Issue #18) improved this document substantially by elevating `G1`–`G6` into non-expert product selling points and explicitly bounding our scope away from political opinion and satire.

Calvin Salsali

1. **What went well:** Framing the problem around the everyday browser user's experience made it clear why a browser extension is much more valuable than a standalone website requiring copy-paste uploads.
2. **Pain points and resolution:** Initially, our stakeholder section was too brief (Issue #16) and our goals read like test cases (Issue #18). I resolved this by creating three concrete user personas (Maya, Marcus, and Priya) and rewriting `G1`–`G6` as clear value propositions while moving test percentages into the SRS.
3. **Scope adjustment:** We moved the browser extension into our core scope (since in-page selection is central to user value) while excluding audio/video deepfakes and subjective opinion/satire scoring (Issue #15).

Raymond Tao

1. **What went well:** Identifying the dual risk of false negatives (falling for scams/synthetic media) and false positives (unfairly distrusting human or ESL authors) gave our project a strong scientific motivation.
2. **Pain points and resolution:** Team 12 rightly asked why we chose 0.80 F_1 and 10% FPR (Issue #14). We resolved this by moving the mathematical formalization and justification of Macro-F_1, FPR, and +0.10 baseline improvement into the SRS while keeping `G3` focused on low-false-accusation quality.
3. **Scope adjustment:** Rather than training foundation models from scratch, we scoped our ML work to calibrating and benchmarking pretrained open-weight detectors (RoBERTa/DeBERTa and ViT/CLIP + EXIF/C2PA) on held-out corpora (RAID, HC3, CIFAKE, GenImage) with an explicit POC fallback rule.

Jaden Moore

1. **What went well:** Defining high-level inputs and outputs early allowed us to see that "insufficient evidence" (`inconclusive`) and "service error" (`failed`) must be first-class outputs rather than hidden exceptions.

2. **Pain points and resolution:** In our first draft, the Environment section repeated usage methodology instead of focusing on the runtime environment (Issue #18). We streamlined Section 1.4 to focus strictly on desktop Chromium browsers and zero client hardware requirements.

3. **Scope adjustment:** We scoped out user accounts and persistent cloud history so the backend service can remain stateless, fast, and manageable within a four-person two-term capstone.

Richard Han

1. **What went well:** Emphasizing privacy-first ephemeral processing (`G6`) and non-accusatory framing (`G4`) from day one aligned our Problem Statement directly with our Hazard Analysis and SRS safety requirements.

2. **Pain points and resolution:** Early drafts did not sufficiently highlight how false positives harm content creators and ESL writers (Issues #14, #16). We resolved this by adding Content Creators, ESL Authors, and Depicted Persons as a core stakeholder group in Table 2.

3. **Scope adjustment:** By bounding our verification claims to observable statistical and provenance signals rather than claiming universal truth arbitration, we ensured every product goal can be verified objectively in our V&V Plan.

References

[1] C. Salsali, R. Tao, J. Moore, and R. Han, "Software Requirements Specification for Verity," Team 23 Capstone Documentation, McMaster University, Oct. 2026.